Supplementary File.

Code snippet: The computation methodology is illustrated in the following snippet:

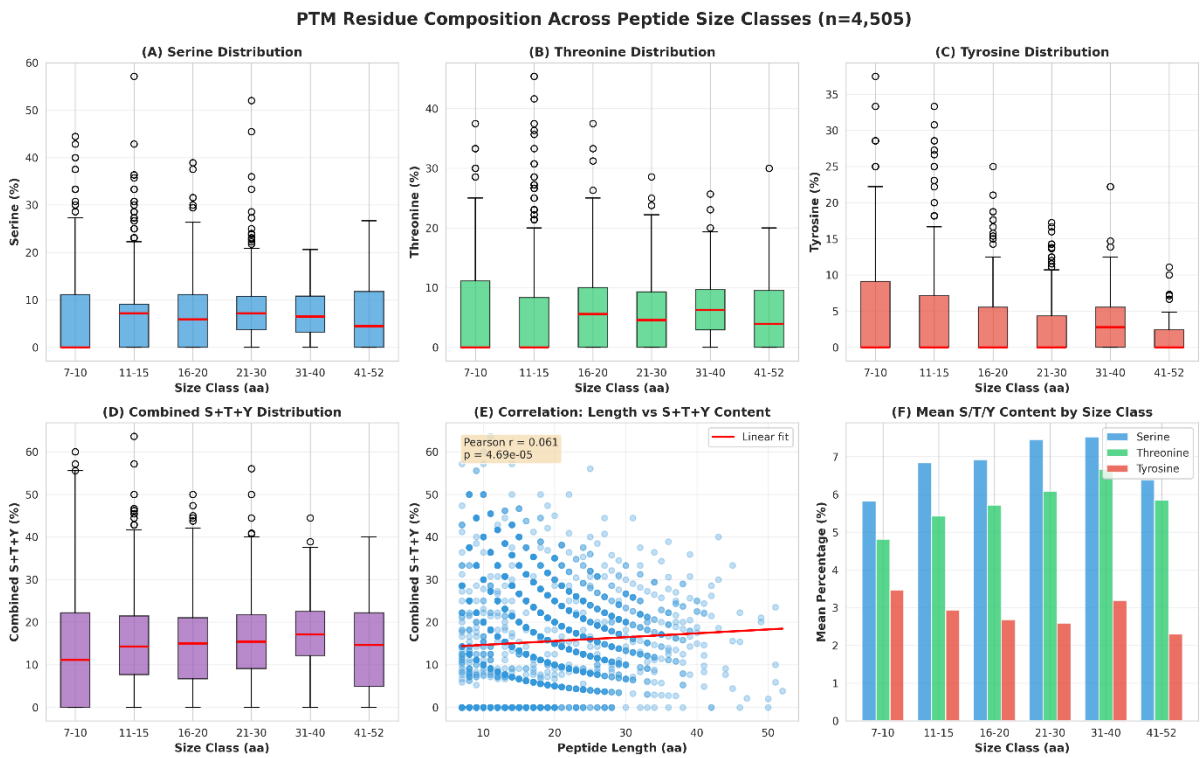
```
#importing the required biopython packages
from Bio.Seq import Seq
from Bio.SeqUtils import ProtParam

#reading the sequence file stored in csv format
df = pd.read_csv("seq.csv")

#defining the function to calculate the amino acid frequency/composition
def calculate_aa_frequencies(sequence):
    analysis = ProtParam.ProteinAnalysis(str(sequence))
    aa_frequencies = analysis.get_amino_acids_percent()
    return pd.Series(aa_frequencies)

#including the columns for each amino acid residue
aa_columns = ['A', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'K', 'L', 'M', 'N', 'P',
'Q', 'R', 'S', 'T', 'V', 'W', 'Y']
#computing the amino acid composition and concatenating the result into the
source file
aa_frequencies_df = df['sequence'].apply(calculate_aa_frequencies)
df = pd.concat([df, aa_frequencies_df], axis=1)
df.to_csv("seq_freq.csv", index=False)
```

Supplementary Figure 1



Supplementary Figure S1: PTM residue composition analysis across peptide size classes. (A–C) Box plots showing the distribution of serine, threonine, and tyrosine percentages across six size classes (n = 4,505 unique peptides). Red lines indicate median values. (D) Combined S+T+Y content distribution by size class. (E) Scatter plot illustrating the correlation between peptide length and combined S+T+Y percentage, with linear regression fit (Pearson $r = 0.061$, $p = 4.69 \times 10^{-5}$). (F) Mean percentage of S, T, and Y residues in each size class, showing progressive increase in serine and threonine content with peptide length. Statistical analysis revealed significant differences in S+T+Y content across size classes (Kruskal-Wallis $H = 44.98$, $p = 1.46 \times 10^{-8}$).

Supplementary Table 1: Distribution of the number of peptide sequences by length

Length of peptide sequence	Number of peptides
>30	171
20-30	733
10-20	2834
<10	767

Supplementary Table 2: PTM-prone residue composition across peptide size classes

Size Class (aa)	n	Serine (%)	Threonine (%)	Tyrosine (%)	Combined S+T+Y (%)
7–10	1102	5.82 ± 8.34	4.82 ± 7.12	3.46 ± 6.24	14.09 ± 12.15
11–15	1582	6.84 ± 7.59	5.43 ± 6.86	2.93 ± 4.98	15.19 ± 10.89
16–20	946	6.92 ± 6.70	5.72 ± 5.71	2.68 ± 4.18	15.32 ± 9.33
21–30	663	7.45 ± 6.40	6.08 ± 5.44	2.58 ± 3.60	16.11 ± 9.01
31–40	179	7.52 ± 5.29	6.66 ± 5.31	3.19 ± 3.83	17.37 ± 8.40
41–52	33	6.39 ± 6.99	5.84 ± 6.86	2.29 ± 3.54	14.52 ± 10.85

Supplementary Table 3: Sequence, length, and serine, threonine and tyrosine contents of peptides with high-confidence structures

Sequence	Length	S	T	Y
IYYINSAASFFAFIATLLYILHAFSIYYH	29	10.34	3.45	17.24
LLDPRIYYINSAASFFAFIATLLYILHAFSIYYH	34	8.82	2.94	14.71
ATSRYEPVAEIGVGAYGTVYKARD	24	4.17	8.33	12.50
PGCYDNGKHYQINQQWERTYLGNALVCTCYGGSR	34	2.94	5.88	11.76

NYGSYSTQASAAAATAELLK	20	15.00	10.00	10.00
PRPGVTEATITGLEPGTEYTIYVIALK	27	0.00	18.52	7.41
YEKPGSPPREVVPRPRPGVTEATITGLEPGTEYTIYVIALK	41	2.44	12.20	7.32
MQYGGAPGGPAFPGQTQDPLYGYFAAVAGQDGQIDADELQR	41	0.00	2.44	7.32
EVTSDSGSIVVSGLTPGVEYVYTIQVLR	28	14.29	10.71	7.14
NQAFLELATEEAAITMVNYYSAVTPHLR	28	3.57	10.71	7.14
PRPGVTEATITGLEPGTEYTIYVIALKNNQK	31	0.00	16.13	6.45
GQSALVGSSYMSALEVGGLVGSIAAGYLSDR	31	19.35	0.00	6.45
EVVPRPRPGVTEATITGLEPGTEYTIYVIALK	32	0.00	15.63	6.25
EVTSDSGSIVVSGLTPGVEYVYTIQVLRDGQER	33	12.12	9.09	6.06
LTIVDTPGFGDAVNNTECWKPITDYVDQQFEQYFR	35	0.00	11.43	5.71
PGVTEATITGLEPGTEYTIYVIALKNNQKSEPLIGR	36	2.78	13.89	5.56
ATFYTPQNPSSLAPAPLRPLEDREQLYSTILSHQCQR	37	10.81	8.11	5.41
CFDHAAGTSYVVGETWEKPYQGWMMVDCTCLGEGSGR	37	5.41	8.11	5.41
GFTINEYTIRPLGVTGVAGEPLPVDSEKDIFDYIQWK	37	2.70	8.11	5.41
AITTGLQEYVEAVSFQHFIK	20	5.00	10.00	5.00
RYNEDLELEDAIHTAILTLK	20	0.00	10.00	5.00
EVTSDSGSIVVSGLTPGVEYVYTIQVLRDGQERDAPIVNK	40	10.00	7.50	5.00
TQEEGEERAESEQEAYLRED	20	5.00	5.00	5.00
REMEENFAVEAANYQDTIGR	20	0.00	5.00	5.00
SSDFEGYEQDFAVLTAEITSK	21	14.29	9.52	4.76
QVTITGSAASISLAQYLINVR	21	14.29	9.52	4.76
QLFELLCYAESIDDQLNTLLK	21	4.76	4.76	4.76
ADRDESSPYAAMLAAQDVAQR	21	9.52	0.00	4.76
ACPLDQAIGLLVAIFHKYSGR	21	4.76	0.00	4.76
GFGDLKSPAGLQVLNDYLADK	21	4.76	0.00	4.76
SASVVSVISRFLEEYLSSTPQR	22	27.27	4.55	4.55
FACNGTVIEHPEYGEVIQLQGDQR	24	0.00	4.17	4.17
FACNGTVIEHPEYGEVIQLQGDQRK	25	0.00	4.00	4.00
DSIVQLCTARPERPMAFLREYFERLEK	27	3.70	3.70	3.70
LKSENQVLGQYIENLMSASSVFQTTDTK	28	14.29	10.71	3.57
EAITAATCLTEALVDHLNVGVAQAYMNQR	29	0.00	10.34	3.45
GIQKFPGINYPVLTPNLKGFEAAVAAGAK	29	0.00	3.45	3.45
AGSVPSLAAGLLFGSLAGLGAYQLSQDPR	29	13.79	0.00	3.45
RPHETGGYMLECVCLGNGKGEWTCKPIAEK	30	0.00	6.67	3.33
AYQDMDIDKVDELMTDITEQQEVAQQISDAISR	33	6.06	6.06	3.03
TGLDSPTGIDFSDITANSFTVHWIAPRATITGYR	34	8.82	17.65	2.94
LSWSVAQGPFDSFVVQYEDTNGQPQALLVDGDQSK	35	11.43	2.86	2.86
TGLDSPTGIDFSDITANSFTVHWIAPRATITGYRIR	36	8.33	16.67	2.78
YETLCQELALITQAVQSEMLETIPEEAGITMSPGPQ	36	5.56	11.11	2.78
IGDTWRRPHETGGYMLECVCLGNGKGEWTCKPIAEK	36	0.00	8.33	2.78
GDSPASSKPISINYRTEIDKPSQMQVTDVQDNSISVK	37	18.92	5.41	2.70
QSVFPFESGKPFKIQVLVEPDHFKVAVNDAHLLQYNHR	38	5.26	0.00	2.63
NSITLTNLTPGTEYVVSIVALNGREESPLLIGQQSTVSDVPR	42	11.90	11.90	2.38
MQIFVKTLTGKTITLEVEPSDTIENVK	27	3.70	18.52	0.00
FGVVQTTLPAILNTLLTLQEAVDK	24	0.00	16.67	0.00

TVFQKENGTVTAANASTLNDGAAALVLMTADAAK	34	2.94	14.71	0.00
VVTPLSPPTNLHLEANPDTGVLTVSWER	28	7.14	14.29	0.00
DITDTLVAVTISEGAHHLDLR	21	4.76	14.29	0.00
TVFQKENGTVTAANASTLNDGAAALVLMTADAAKR	35	2.86	14.29	0.00
QKTGLDSPTGIDFSDITANSFTVHWIAPR	29	10.34	13.79	0.00
ARDMDLTEVITGTLWNLSSHDSIK	24	12.50	12.50	0.00
DLEVVAATPTSLLISWDAPAVTVR	24	8.33	12.50	0.00
DDKESVPISDTIIPAVPPPTDLRFTNIGPDTMR	33	6.06	12.12	0.00
DAPIVNKVVTPLSPPTNLHLEANPDTGVLTVSWER	35	5.71	11.43	0.00
DGQERDAPIVNKVVTPLSPPTNLHLEANPDTGVLTVSWER	40	5.00	10.00	0.00
EFETAETLLNSEVHMLLEHR	20	5.00	10.00	0.00
EENVGLHQTLDQTLNELNCI	20	0.00	10.00	0.00
RVLLPTRPAPPTVEQILEDVR	21	0.00	9.52	0.00
TIDDLEEKLAQAKEENVGLHQTLDQTLNELNCI	33	0.00	9.09	0.00
NTTLGNSLQESLDELIQSQQITPQLALQVLLQFDK	35	8.57	8.57	0.00
IKLIDMLSEAGLSVIETTSFVSPK	24	16.67	8.33	0.00
AANATTNPSQLLPLELVDKCIGSR	24	8.33	8.33	0.00
DSMIWDCTCIGAGRGRISCTIANR	24	8.33	8.33	0.00
AVENPTATEIQDVCSAVGLNVFLEK	25	4.00	8.00	0.00
SSPVVIDASTAIDAPSNLRFLATTPNSLLVSWQPPRAR	38	15.79	7.89	0.00
IHQMEENQHSTSEELQATLQELADLQQITQELNSENER	38	7.89	7.89	0.00
LGLQLLELPPEESLPLGPLLGDTAVIQGDTALITRPWSPAR	41	4.88	7.32	0.00
STNLLAAETASLEEQLQGWGEVMLMADK	28	7.14	7.14	0.00
ITNDCPEHLESIDVMCQVLTDLIDEEVK	28	3.57	7.14	0.00
TPIGSFLGSLSLLPATKLGSIAIQGAIEK	29	13.79	6.90	0.00
TGACQDVDECQAIPGLCQGGNCINTVGSFECK	32	3.13	6.25	0.00
NLPGGQQNSSWNFSEDTVISILNTINEVIAENLEAAK	37	10.81	5.41	0.00
LSKPELLTLFSILEGELEAR	20	10.00	5.00	0.00
EENLSMHQMLDQTLLELNNM	20	5.00	5.00	0.00
FCGDLDCPDWVLAEISTLAK	20	5.00	5.00	0.00
GAPPKIPGGATLVFEVELLK	20	0.00	5.00	0.00
LIETLKPFGVFEEEEELQRR	20	0.00	5.00	0.00
ECVQECVSEFISFITSEASER	21	19.05	4.76	0.00
DSLSVNEFKELVTQQLPHLLK	21	9.52	4.76	0.00
AAELFHQLSQALEVLTDAAAR	21	4.76	4.76	0.00
NWLLFACHATNEVAQLIQGGR	21	0.00	4.76	0.00
IGEKWDRQGENGQMMSCTCLGNGK	24	4.17	4.17	0.00
GPRGETGPAGRPGEVGPPGPPGPAGEKGSPGADGPAGAPGTPGPQGIAGQR	51	1.96	3.92	0.00
QGAPGQGGGGGLSHEDTLALLEGLVSR	27	7.41	3.70	0.00
RLNTILAQIMPFLSQEHQQQVAQAVER	27	3.70	3.70	0.00
LRPQSAQSSFPPSPGPSPDVQLATLAQR	28	17.86	3.57	0.00
SDEHVVSGNLVTPLPVILGHEAAGIVESV	29	10.34	3.45	0.00
QFQADLQTAVVIANDIKSEAQEEIGDLKR	29	3.45	3.45	0.00
GDRGETGPAGPPGAPGAPGAPGPVGPAGK	29	0.00	3.45	0.00
LPSVPSTHLPAGPAPKVDEDEEALKQLAEWVS	32	9.38	3.13	0.00
SIDDLEEKVAHAKEENLSMHQMLDQTLLELNNM	33	6.06	3.03	0.00

SWFEPLVEDMQRQWAGLVEKVQAAVGTSAAPVPSDNH	37	8.11	2.70	0.00
LHGSVGGAQNLSALGALVSLSNAR	24	16.67	0.00	0.00
NFDSLESLISAIQGDIEEAKK	21	14.29	0.00	0.00
SQDGASQFQEVIRQELELSVK	21	14.29	0.00	0.00
GISLNPEQWSQLKEQISDIDDAVR	24	12.50	0.00	0.00
SGLAIVSQHDLDQLQADAINAFGESLQK	28	10.71	0.00	0.00
QLLSNPEMMVQIMENPFVQSMLSNPDLMR	29	10.34	0.00	0.00
SGLAIVSQHDLDQLQADAINAFGESLQKK	29	10.34	0.00	0.00
GPQLAAQNLGISLANLLLSK	20	10.00	0.00	0.00
LESGHLPSMQQLVQILDNEMK	21	9.52	0.00	0.00
AEDSLLAAEEAAAKAEADVASLNR	24	8.33	0.00	0.00
ISQAEEEDQQLLGHLLLVAK	20	5.00	0.00	0.00
VVLPMPILVLPPIVMSMLEK	20	5.00	0.00	0.00
AAPSGGVNCEEFAEFQELLK	20	5.00	0.00	0.00
ILVSCNNQQDLQEWVEHLQK	20	5.00	0.00	0.00
LVILEGELERAEERAEVSELK	21	4.76	0.00	0.00
EDIAMWEVNEAFSLVVLANIK	21	4.76	0.00	0.00
FIPLSEPAPVPPIPNEQQLAR	21	4.76	0.00	0.00
GEQGPAGSPGFQGLPGPAGPPGEAGKPGEQGVPGDLGAPGPSGAR	45	4.44	0.00	0.00
DVGLKKEDIAMWEVNEAFSLVVLANIK	27	3.70	0.00	0.00
GLPGPPGAPGPQGFQGPPGEPGEPGASGPMGPRGPPGPPGKNGDDGEAGK	50	2.00	0.00	0.00